

Red Hat Lightspeed 1-latest

Registering RHEL systems and configuring client tools with Red Hat Lightspeed with FedRAMP

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Last Updated: 2026-09-02

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Abstract

This document provides information about how to register your RHEL systems and configure Red Hat Lightspeed with client tools with FedRAMP®.

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CHAPTER 1. CLIENT TOOLS AND COMPONENTS

RHEL client tools such as insights-client and subscription-manager handle system registration, entitlement management, and proactive remediation. These tools integrate with each other to help you connect your RHEL system to Red Hat services.

1.1. CLIENT TOOLS AND COMPONENTS OVERVIEW

Red Hat Enterprise Linux (RHEL) includes several client tools that work together to register your systems and connect them to Red Hat services.

subscription-manager

Handles system identity and subscription entitlements, and provides access to Red Hat package repositories through the Content Delivery Network (CDN).

insights-client

Gathers system data and sends it to Red Hat Lightspeed for analysis. It depends on subscription-manager for system identity.

1.2. THE SUBSCRIPTION-MANAGER CLIENT

The subscription-manager client establishes your system's identity and provides access to the Red Hat Content Delivery Network (CDN) for updates and packages. This registration method provides limited Red Hat Subscription Management features.

1.3. THE INSIGHTS-CLIENT

The insights-client collects system data and sends it to Red Hat Lightspeed for analysis. The insights-client relies on the subscription-manager client to establish the identity of the system.

The insights-client is available for these RHEL releases.

Table 1.1. RHEL releases with insights-client availability

RHEL release	Comments
RHEL 10	Distributed with insights-client pre-installed.
RHEL 9	Distributed with insights-client pre-installed.
RHEL 8	Distributed with insights-client pre-installed, unless RHEL 8 was installed as a minimal installation.
RHEL 7	Distributed with the insights-client RPM package loaded but not installed.

Additional resources

- Getting started with Red Hat Lightspeed with FedRAMP

1.4. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

1.4.1. User Access overview

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

All users on your account have access to most of the data in Red Hat Lightspeed.

Additional resources

- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP

1.4.2. Predefined groups in User Access

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, the group is automatically renamed to **Custom default access** Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

1.4.3. Predefined roles assigned to groups

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

TIP

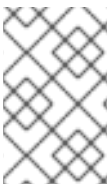
Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Predefined User Access roles

1.4.4. Check your permissions

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.

2. If you try to access Red Hat Lightspeed features and see a message that you do not have permission to perform this action, contact your Organization Administrator or a user with the **User Access administrator** role to request the permissions required to access those features and complete the actions you want to perform.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

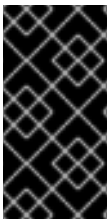
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

1.4.5. Configure user permissions

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features
- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization

- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

1.5. INSTALL THE INSIGHTS-CLIENT

The insights-client is automatically installed with default RHEL installations. Manual installation is required for deployments such as minimal and cloud-based installations.

Installation steps vary slightly depending on your RHEL version and installation type. After you install the insights-client, register your system. Some registration methods require activation keys.

Additional resources

- Managing Activation Keys in Red Hat Satellite

1.6. CLIENT COMMAND-LINE INTERFACE AND CONFIGURATION INTERACTION

The insights-client runs automatically based on its scheduler settings. By default, it runs every 24 hours. To run the client interactively, enter the **insights-client** command.

When you run **insights-client**, the following values and settings determine the results:

1. Values that you enter when you run **insights-client** from the command line temporarily override the preset configuration file settings and system environment settings. Any values that you enter for options in the **insights-client** command are used only for that instance of insights-client.
2. Settings in the configuration file (**/etc/insights-client/insights-client.conf**) override system environment settings.
3. Values of any system environment variables (**printenv**) are not affected by the commands entered on the command line or the client configuration files.

CHAPTER 2. CONNECT AND REGISTER SYSTEMS TO RED HAT SERVICES

Register your RHEL system to Red Hat services by using subscription-manager and insights-client.

2.1. REGISTER RHEL SYSTEMS

Using Red Hat Lightspeed with FedRAMP requires configuring your systems to communicate with Red Hat using Red Hat Satellite. After configuring your system to communicate with Satellite, you can use insights-client and subscription-manager to register your systems. For more information, see Registration methods, or contact Red Hat support for assistance.

- Register to Red Hat Subscription Manager (RHSM) with the subscription-manager client.
- Register the system with insights-client.

Prerequisites

- You have access to the system as root or have sudo permissions.
- You have an activation key and an organization ID to register your system.
- You have completed the steps to Manage system connectivity and networking .
- Optional: You have completed steps to Minimize data collection and control data privacy .

Procedure

1. To register a system running Red Hat Enterprise Linux version 7 or 8, use an activation key and your Organization ID to register with RHSM.

```
# subscription-manager register --activationkey=_activation_key_name_ --
org=_organization_ID_
```

2. Enter the **insights-client** command to complete registration for the Red Hat Lightspeed service.

```
# insights-client --register
```

Additional resources

- Getting started with Red Hat Lightspeed with FedRAMP
- Getting started with activation keys on the Red Hat Hybrid Cloud Console
- Getting started with RHEL System Registration

CHAPTER 3. UNREGISTER SYSTEMS

To stop uploading data to Red Hat services, unregister your systems. You can use the insights-client to disconnect specifically from Red Hat Lightspeed.

3.1. UNREGISTER YOUR SYSTEM WITH RED HAT LIGHTSPEED

You can unregister your system with Red Hat Lightspeed. When you do so, your system information is no longer uploaded to Red Hat Lightspeed.

Prerequisites

- Root-level access to your system.
- Your system is registered with Red Hat Lightspeed.

Procedure

- Enter the **insights-client** command with the **--unregister** option.

```
[root@rhlightspeed]# insights-client --unregister
```

Verification

- Enter the following command to confirm the status and verify that the system has been successfully unregistered:

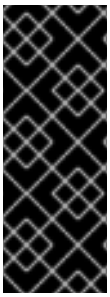
```
[root@rhlightspeed]# insights-client --status
```

If the command is successful, you will see the following output:

```
This host is unregistered.
```

3.2. RE-REGISTER YOUR SYSTEM WITH RED HAT LIGHTSPEED

Re-registration is not recommended but is sometimes necessary for changing feature levels or rebuilding systems. Use the insights-client to unregister and re-register to avoid duplicate inventory entries.



IMPORTANT

If you are a Red Hat Satellite user and you plan to upgrade or rebuild your Satellite system by doing a fresh install, you must re-register that system in Red Hat Lightspeed. Ensure that you unregister the system in Red Hat Lightspeed before you begin the upgrade. After you have reinstalled the Satellite system, register it again with the **insights-client**. Otherwise, you might see duplicate host entries or other unexpected results.

To re-register a system in Red Hat Lightspeed, use the **insights-client --unregister** and **--register** commands.

Prerequisites

You have completed the following steps on your system:

- Logged in with root-level permissions
- Installed the insights-client

Procedure

1. On the command line, enter the **insights-client** command with the **--unregister** option.

```
[root@rhlightspeed]# insights-client --unregister
```

2. Enter the **insights-client** command with the **--register** option.

```
[root@rhlightspeed]# insights-client --register
```

Verification

- Successfully re-registering a system by using the **insights-client** command, creates a new Red Hat Lightspeed profile, and results in the display of following output in your terminal:

```
[root@rhlightspeed]# Successfully uploaded report for <machine name>
View the Red Hat Lightspeed console at https://console.redhat.com/insights/
```

CHAPTER 4. MANAGE SYSTEM OPERATIONS AND CONFIGURATION SETTINGS

Manage system connectivity and client settings by using insights-client.

4.1. SYSTEM CONNECTIVITY AND NETWORKING

Configure network and firewall settings to connect systems to Red Hat Lightspeed through Red Hat Satellite.

Your systems can connect to Red Hat Lightspeed using Red Hat Satellite servers. The networking requirements differ based on your architecture. For Satellite-managed systems, the Satellite server handles routing to Red Hat services for content delivery and data uploads.

Additional resources

- Red Hat Satellite documentation
- Red Hat Customer Support

4.2. SYSTEM DISPLAY AND STATUS

Manage system identity and connection status to maintain accurate inventory and prevent duplicate entries that can lead to inaccurate reporting.

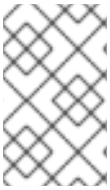
4.2.1. Change the host display name

Use insights-client to set a custom display name during or after registration to improve inventory identification. If you do not assign a display name, Red Hat Lightspeed uses the value in **/etc/hostname**.



NOTE

Using the **insights-client** command to set the display name takes effect immediately, but does not run the client.



NOTE

If you obfuscate the hostname, the **hostname** configured in **/etc/hostname** is obfuscated. Assign a **display name** so that you can identify a host even when its **hostname** is obfuscated.

Prerequisites

- Root-level access to the system.

Procedure

- Enter the **insights-client** command with the **--display-name** option and specify a display name.

```
[root@rhlightspeed]# insights-client --display-name ITC-4
System display name changed from None to ITC-4
```

- To create a display name that contains spaces, use double quotes.

```
[root@rhlightspeed]# insights-client --display-name "ITC-4"
System display name changed from None to ITC-4
```

Additional resources

- Assigned system display names

4.2.2. Display the client version

You can display the insights-client version and client core version.

Prerequisites

- Root-level access to your system.

Procedure

- Enter the **insights-client** command with the **--version** option.

```
[root@rhlightspeed]# insights-client --version
Client: 3.0.6-0
Core: 3.0.121-1
```

Additional resources

- Red Hat insights-client Core Changelog (Red Hat Knowledgebase)
- Changelog file

4.3. TIMER SCHEDULE AND RULE UPDATES FOR THE INSIGHTS-CLIENT

Control when insights-client runs by disabling, enabling, or modifying the default 24-hour schedule.

By default, the insights-client runs every 24 hours. The timers in the default schedules are random so that all systems do not run the client at the same time. You can disable, enable, or modify the schedule to control when the system performs data collection, although changes from the default might affect performance. To ensure that host information remains up-to-date, extending the timers beyond 24 hours is not recommended.

4.3.1. Disable the insights-client schedule

You must disable the client schedule before you can change the default insights-client settings and create a new schedule. The procedure you use to disable the insights-client schedule depends on your Red Hat Enterprise Linux and client versions.

Prerequisites

- Root-level access to your system
- The insights-client version 3.x and later

- NOTE: The **--no-schedule** option is deprecated in Client 3.x and later.

Procedure

1. Enter the **insights-client** command with the **--version** option to verify that you have the required client version installed.

```
[root@rhlightspeed]# insights-client --version
Client: 3.0.6-0
Core: 3.0.121-1
```

2. Enter the **insights-client** command with the **--disable-schedule** option to disable the client schedule.

```
[root@rhlightspeed]# insights-client --disable-schedule
```

4.3.2. Enable the insights-client schedule

When you enable the client schedule, it runs using its default settings. If you change the schedule, those settings take precedence.

When you run **insights-client** from the command line, insights-client runs using the settings you specify for only that session. When the next scheduled run takes place, it uses the default settings.

Prerequisites

- Root-level access to your system.
- The client schedule is disabled.
- Optional: You modified the default schedule.

Procedure

1. Verify the client version on your system by entering the **insights-client** command with the **--version** option.

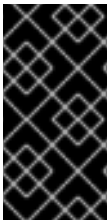
```
[root@rhlightspeed]# insights-client --version
Client: 3.0.6-0
Core: 3.0.121-1
```

2. Enter the **insights-client** command with the **--enable-schedule** option to enable the client schedule.

```
[root@rhlightspeed]# insights-client --enable-schedule
```

4.3.3. Modify the insights-client schedule by using the systemd service

You can change when the insights-client runs by modifying the schedule. The method that you use depends on the RHEL release and client version that your system is running. For Red Hat Enterprise Linux 7.5 and later, you can update the **systemd** settings and the **insights-client.timer** file.



IMPORTANT

If you update the schedule using **cron** or the **systemd.timer**, those customized settings take precedence. However, if you change settings using the insights-client on the command line, the settings apply only to that session. When the next scheduled run takes place, the client uses your persistent customized settings.



NOTE

For Red Hat Enterprise Linux 7.4 and earlier, use **cron** to modify the system schedule. For more information about using **cron**, see the Red Hat Knowledgebase solution, What is **cron** and how is it used? .

Prerequisites

- You have root-user access to your system.

Procedure

1. To edit the settings in the **insights-client.timer** file, enter the **systemctl edit** command and the file name.

```
[root@rhlightspeed]# systemctl edit insights-client.timer
```

   This action opens an empty file with the default system editor, on RHEL 6, 7, and 8. On RHEL 9 and later, running the command opens a file with a template that includes the original **systemd** service file as a reference.

2. Enter different settings to modify the schedule. The values in this example are the default settings for **systemd**.

```
[Timer]
OnCalendar=daily
RandomizedDelaySec=14400
```

3. Enable the **insights-client** schedule.

```
[root@rhlightspeed]# insights-client --enable-schedule
```

   - For more information, refer to the man pages for **systemctl(1)**, **systemd.timer(5)**, and **systemd.time(7)** to understand **systemd**

4.3.4. Enable automatic rule updates for Red Hat Lightspeed

Re-enable the settings for automatic collection rule updates in the insights-client configuration to keep collection rules current. Automatic rule collections helps you benefit from the latest security and performance analyses provided by Red Hat Lightspeed.

Prerequisites

- Root-level access to your system.
- Automatic rule collection is disabled.

Procedure

1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the line that contains the following:

```
auto_update=False
```

3. Change **False** to **True**.

```
auto_update=True
```

4. Save and close the **/etc/insights-client/insights-client.conf** file.

4.3.5. Disable automatic rule updates

Disable automatic collection rule updates in the insights-client configuration. Automatic updates keep collection rules current, so disabling them is not recommended because you do not receive the most recent collection rule updates from Red Hat's hosted services.

Prerequisites

- Root-level access to your system.
- Automatic rule updates are enabled.

Procedure

1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the line that contains

```
#auto_update=True
```

3. Remove the **#** and change **True** to **False**.

```
auto_update=False
```

4. Save and close the **/etc/insights-client/insights-client.conf** file.

4.3.6. Package cache refresh for systems managed by Red Hat Satellite

You can use the **--build-packagecache** option to provide accurate reporting for applicable updates on Satellite-managed systems. This option rebuilds the **yum**/**dnf** package caches for insights-client, and creates a refreshed list of applicable updates for the system.

Run the command manually to rebuild the package caches immediately. Alternatively, edit **/etc/insights-client/insights-client.conf** to rebuild the package caches automatically each time the system checks in to Red Hat Lightspeed.

Additional resources

- Managing system content and patch updates on RHEL systems with FedRAMP

- Red Hat Lightspeed shows incorrect patch reporting for Satellite-managed systems (Red Hat Knowledgebase)
- Managing errata

CHAPTER 5. MINIMIZE DATA COLLECTION AND CONTROL DATA PRIVACY

Configure data obfuscation and redaction to minimize data collection and enhance data privacy for RHEL systems. Obfuscate sensitive identifiers such as IP addresses and hostnames, exclude file content or command output, and prevent exposure of personally identifiable information (PII).

5.1. SYSTEM FACTS

A *data collector* is an application or service that regularly sends host information, updates, or system profile data to Red Hat Lightspeed inventory. **insights-client** is a data collector for Red Hat Lightspeed.

System facts are the metadata that data collectors collect about your RHEL systems. These facts describe runtime configuration, system health, and system performance. The insights-client uses the system facts it collects to populate inventory data in Red Hat Lightspeed and to update existing data. Red Hat Lightspeed also uses system facts to analyze system performance and to create recommendations for services such as advisor or remediations.

Additional resources

- Which RHEL system facts are collected and populated by insights-client (Red Hat Lightspeed)?
- Data collectors for inventory

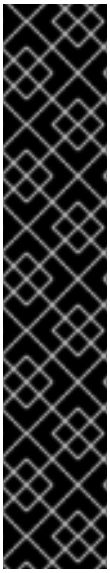
5.2. DATA OBFUSCATION

Red Hat Lightspeed provides optional controls to obfuscate IP addresses, MAC addresses, and hostnames in the data archive and the user interface. These controls mask sensitive information before it reaches Red Hat.

5.2.1. Data obfuscation considerations and limitations

Obfuscation capabilities depend on your connection type and client tool configuration. Consider feature level requirements and data privacy when configuring obfuscation.

Several client tools and products from Red Hat work together to provide a full range of functionalities. Depending on your setup, different tools can collect information from your systems at the same time. Some of the tools have built-in features for configuring obfuscation or redaction. To achieve the required state of data privacy, you must take many aspects into consideration, such as the feature-level requirements and how the system consumes the content.



IMPORTANT

Directly connected systems. You cannot achieve true obfuscation or redaction of system information. The subscription-manager must remain enabled on the system and continues to report data.

This applies regardless of how insights-client is configured.

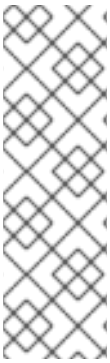
You cannot prevent the collection of certain key system information such as the hostname, or IP and MAC addresses. For directly connected systems, even configuring data obfuscation in the insights-client does not achieve the expected results.

Systems connected to a Red Hat Satellite or Capsule. You cannot fully prevent collection of key system information such as hostname, IP, and MAC addresses. However, you can adjust the feature level and configure Satellite so that data is never sent to Red Hat.

5.2.2. Data obfuscation on the system

Full data obfuscation requires systems that are registered to Red Hat Satellite with client tools and server settings configured to enable obfuscation.

You can enable the insights-client obfuscation feature to replace the hostname, IP address, or MAC address with preset values in the Red Hat Lightspeed archive. A Python data cleaning process generates the replacement values. The processed archive file is then sent to Red Hat Lightspeed. Obfuscation is disabled by default.



NOTE

- The Python data cleaning process automatically generates the masked values. You cannot choose the values for obfuscation.
- The Red Hat Lightspeed compliance service uses OpenSCAP tools to generate compliance reports based on information from the host system. The compliance service uses OpenSCAP, which prevents complete obfuscation or redaction of hostname and IP address data. Host information is also sent to Red Hat Lightspeed when a compliance data collection job runs on the host system.

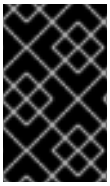
Additional resources

- Red Hat Lightspeed data & application security

5.2.3. Obfuscate IPv4 IP addresses

If your hosts use IPv4 IP addresses, enable obfuscation in the **insights-client** configuration file to mask sensitive data in the archive file before it is sent to Red Hat Lightspeed.

When you enable obfuscation, the original IP address of the host is replaced with a generated value in the archive file. This obfuscated value is used in the Red Hat Hybrid Cloud Console UI, logs, and any archive data files that Red Hat collects. However, you still see the original IP address in the command-line output of some **insights-client** commands.



IMPORTANT

The obfuscation process uses a Python data cleaning process to generate a unique value for each host. You cannot configure the value provided for obfuscation. You also cannot obfuscate or select the portion of the host IP address to obfuscate.

Prerequisites

- If you are using Red Hat Satellite to manage clients and register them on the Red Hat Hybrid Cloud Console, you must also configure the **Red Hat Cloud** settings before you can enable obfuscation in Red Hat Lightspeed:
  - In the Satellite web UI, navigate to **Administer → Settings → Red Hat Cloud** and enable the required obfuscation settings that apply to your environment.

Procedure

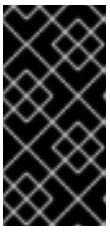
1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the following section:

```
# Specify which will be obfuscated in the data collection,
# empty by default, and supported options are: ipv4, ipv6, hostname, mac
# (comma-separated list)
# obfuscation_list=
```

3. Remove the preceding hash (**#**) character, before **obfuscation_list=**, and add the following line:

```
obfuscation_list=ipv4
```

4. If your configuration file contains the older **obfuscate=True** or **obfuscate=False** Red Hat Lightspeed obfuscation setting, remove that line of configuration.



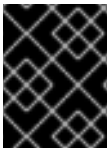
IMPORTANT

Remove deprecated **obfuscate** and **obfuscate_hostname** options from the configuration file. If your configuration file contains both **obfuscation_list** and deprecated options, the **obfuscation_list** takes precedence and insights-client displays a warning message in the output.

5. Save and close the **/etc/insights-client/insights-client.conf** file.

Result

When obfuscation is enabled, the original IPv4 address is masked in the console UI, logs, and archive data files that Red Hat collects.



IMPORTANT

After you enable obfuscation, you continue to see the original IPv4 address in the command-line output of some **insights-client** commands.

- Original host system IP address:

> 192.168.0.24

- Obfuscated host IP address:

> 10.230.230.1

- Obfuscated IPv4 IP address in the Red Hat Hybrid Cloud Console UI:





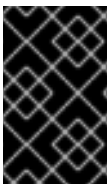
NOTE

When you enable obfuscation on several systems, the same obfuscated IP address gets generated. Therefore, in the example scenario provided, when you search or filter by IP address in the Red Hat Lightspeed UI on the Hybrid Cloud Console you might see several instances of **10.230.230.1**. This is because the Python data cleaning process that the Red Hat Lightspeed obfuscation feature uses, can generate the same obfuscated IP address in the archive file.

5.2.4. Obfuscate IPv6 IP addresses

If your hosts use IPv6 IP addresses, enable obfuscation in the **insights-client** configuration file to mask sensitive data in the archive file before it is sent to Red Hat Lightspeed.

When you enable obfuscation, the original IP address of the host is replaced with a generated value in the archive file. This obfuscated value is used in the Red Hat Hybrid Cloud Console UI, logs, and any archive data files that Red Hat collects. However, you still see the original IP address in the command-line output of some **insights-client** commands.



IMPORTANT

The obfuscation process uses a Python data cleaning process to generate a unique value for each host. You cannot configure the value provided for obfuscation. You also cannot obfuscate or select the portion of the host IP address to obfuscate.

Prerequisites

- If you are using Red Hat Satellite to manage clients and register them on the Red Hat Hybrid Cloud Console, you must also configure the **Red Hat Cloud** settings before you can enable obfuscation in Red Hat Lightspeed:
  - In the Satellite web UI, navigate to **Administer → Settings → Red Hat Cloud** and enable the required obfuscation settings that apply to your environment.

Procedure

1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the following section:

```
# Specify which will be obfuscated in the data collection,
# empty by default, and supported options are: ipv4, ipv6, hostname, mac
# (comma separated list)
# obfuscation_list=
```

3. Remove the preceding hash (**#**) character, before **obfuscation_list=**, and add the following line:

```
obfuscation_list=ipv6
```

4. If your configuration file contains the older **obfuscate=True** or **obfuscate=False** Red Hat Lightspeed obfuscation setting, remove that line of configuration.



IMPORTANT

Remove deprecated **obfuscate** and **obfuscate_hostname** options from the configuration file. If your configuration file contains both **obfuscation_list** and deprecated options, the **obfuscation_list** takes precedence and insights-client displays a warning message in the output.

5. Save and close the **/etc/insights-client/insights-client.conf** file.

Result

When obfuscation is enabled, the original IPv6 address is masked in the console UI, logs, and archive data files that Red Hat collects.



IMPORTANT

After you enable obfuscation, you continue to see the original IP address in the command-line output of some **insights-client** commands.

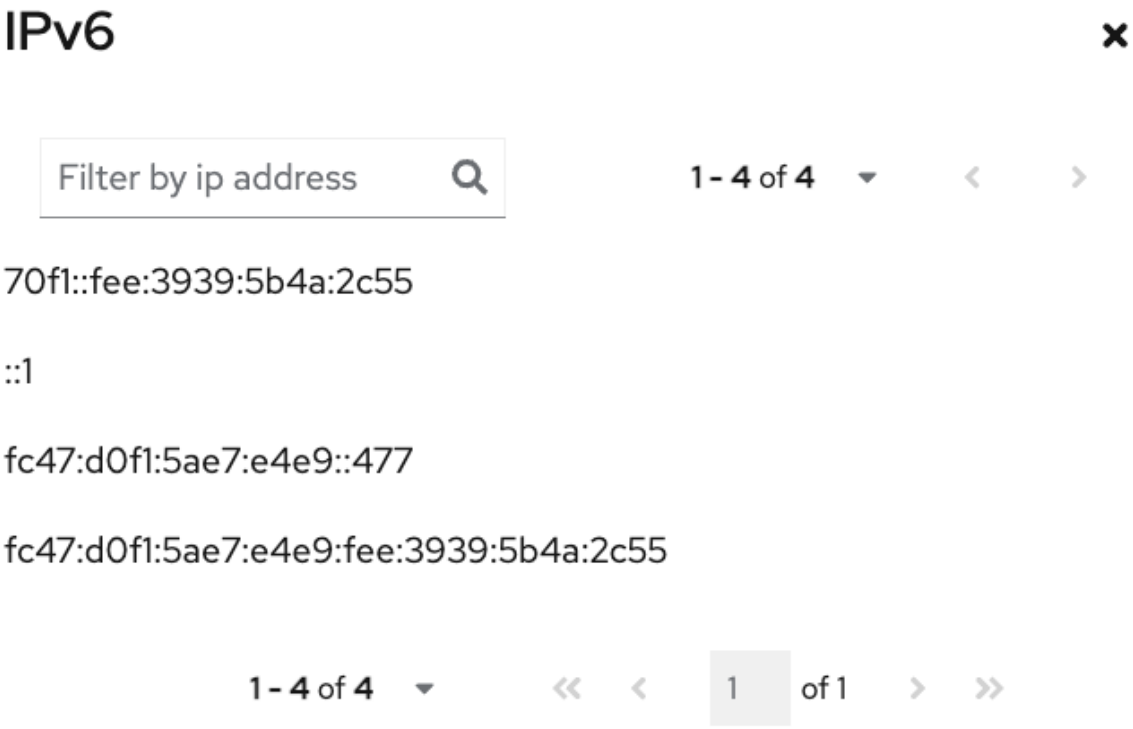
- Original host system IPv6 addresses:

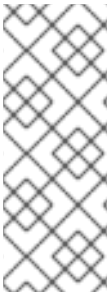
```
ff00:f800:f801:f802::f806
ff00:f800:f801:f802:f00:f803:f804:f805
ff01::f00:f803:f804:f805
```

- Obfuscated host IPv6 addresses:

```
fc47:d0f1:5ae7:e4e9::0477,
fc47:d0f1:5ae7:e4e9:fee:3939:5b4a:2c55,
70f1::fee:3939:5b4a:2c55,
```

- Obfuscated IPv6 IP addresses in the Red Hat Hybrid Cloud Console UI:



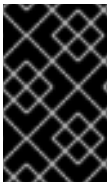


NOTE

When you enable obfuscation on several systems, the same obfuscated IP address gets generated. Therefore, in the scenario provided, when you search or filter by IP address in the Red Hat Lightspeed UI on the Hybrid Cloud Console, you might see several instances of **70f1::fee:3939:5b4a:2c55**. This is because the Python data cleaning process that the Red Hat Lightspeed obfuscation feature uses can generate the same obfuscated IP address in the archive file.

5.2.5. Obfuscate MAC addresses

You can mask the Media Access Control (MAC) addresses of your hosts in the archive file before it is sent to Red Hat Lightspeed by enabling obfuscation.



IMPORTANT

The obfuscation process uses a Python data cleaning process to generate a unique value for each host. You cannot configure the value provided for obfuscation. You also cannot obfuscate or select the portion of the host MAC address to obfuscate.

Prerequisites

- If you are using Red Hat Satellite to manage clients and register them on the Red Hat Hybrid Cloud Console, you must also configure the **Red Hat Cloud** settings before you can enable obfuscation in Red Hat Lightspeed:
  - In the Satellite web UI, navigate to **Administer → Settings → Red Hat Cloud** and enable the required obfuscation settings that apply to your environment.

Procedure

1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the following section:

```
# Specify which will be obfuscated in the data collection,
# empty by default, and supported options are: ipv4, ipv6, hostname, mac
# (comma separated list)
# obfuscation_list=
```

3. Remove the preceding hash (**#**) character, before **obfuscation_list=** and add the following line:

```
obfuscation_list=mac
```

4. If your configuration file contains the older **obfuscate=True** or **obfuscate=False** Red Hat Lightspeed obfuscation setting, remove that line of configuration.



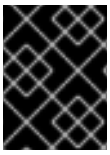
IMPORTANT

Remove deprecated **obfuscate** and **obfuscate_hostname** options from the configuration file. If your configuration file contains both **obfuscation_list** and deprecated options, the **obfuscation_list** takes precedence and insights-client displays a warning message in the output.

5. Save and close the **/etc/insights-client/insights-client.conf** file.

Result

When obfuscation is enabled, the original MAC address is masked in the console UI, logs, and archive data files that Red Hat collects.



IMPORTANT

After you enable obfuscation, you continue to see the original MAC address in the command-line output of some **insights-client** commands.

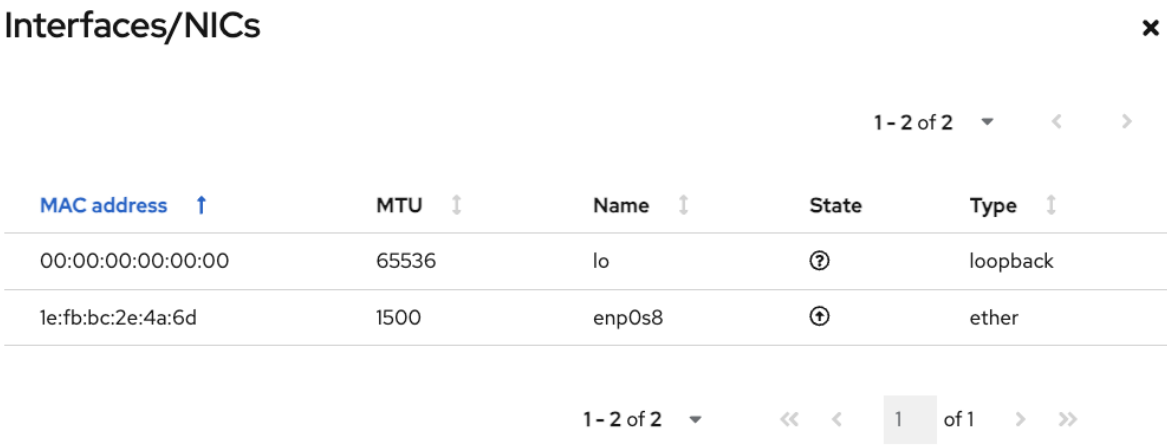
- Original host system MAC address:

```
08:00:27:7c:fc:0f
```

- Obfuscated host MAC address:

```
1e:fb:bc:2e:4a:6d
```

- Obfuscated MAC address in the Red Hat Hybrid Cloud Console:





NOTE

When you enable obfuscation on several systems, the same obfuscated MAC address gets generated. Therefore, in the example scenario provided, when you search or filter by MAC address in the Red Hat Lightspeed UI on the Hybrid Cloud Console, you might see several instances of **1e:fb:bc:2e:4a:6d**. This is because the Python data cleaning process that the Red Hat Lightspeed obfuscation feature uses can generate the same obfuscated MAC address in the archive file.

5.2.6. Obfuscate hostnames

Configure insights-client to obfuscate hostnames by replacing the value in **/etc/hostname** with an automatically generated UUID before sending data to Red Hat.

During data obfuscation, a Python process generates a 12-character UUID to mask the hostname in the data archive file and Hybrid Cloud Console. The masked data is then sent to Red Hat.

TIP

An obfuscated hostname can be difficult to recognize. Setting a display name can help you to more easily identify your obfuscated hosts. The display name does not get obfuscated and displays in the Red Hat Lightspeed console UI. Only the value of **/etc/hostname** gets obfuscated.

Prerequisites

- If you are using Red Hat Satellite to manage clients and register them on the Red Hat Hybrid Cloud Console, you must also configure the **Red Hat Cloud** settings before you can enable obfuscation in Red Hat Lightspeed:
  - In the Satellite web UI, navigate to **Administer → Settings → Red Hat Cloud** and enable the required obfuscation settings that apply to your environment.

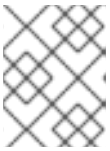
Procedure

1. Open the **/etc/insights-client/insights-client.conf** file with an editor.
2. Locate the following section:

```
obfuscation_list=
```

3. Add the following line to the **obfuscation_list** section to enable hostname obfuscation:

```
obfuscation_list=hostname
```

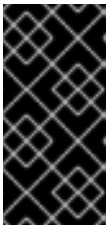


NOTE

To add multiple obfuscation options, separate them with commas. For example, to obfuscate the hostname and an IPv6 IP address, add:

```
obfuscation_list=hostname,ipv6
```

4. If your configuration file contains the older **obfuscate_hostname=True** or **obfuscate_hostname=False** Red Hat Lightspeed obfuscation setting, remove that line of configuration.



IMPORTANT

Remove deprecated **obfuscate** and **obfuscate_hostname** options from the configuration file. If your configuration file contains both **obfuscation_list** and deprecated options, the **obfuscation_list** takes precedence and insights-client displays a warning message in the output.

5. Optional: Assign a display name to your system so that you can more easily find and manage your obfuscated hosts in the Red Hat Lightspeed console UI by adding the following line:

```
display_name=example-display-name
```



NOTE

You can also set a display name by using the following command:

```
[root@rhlightspeed]# insights-client --display-name ITC-4
```

6. Save and close the **/etc/insights-client/insights-client.conf** file.

Result

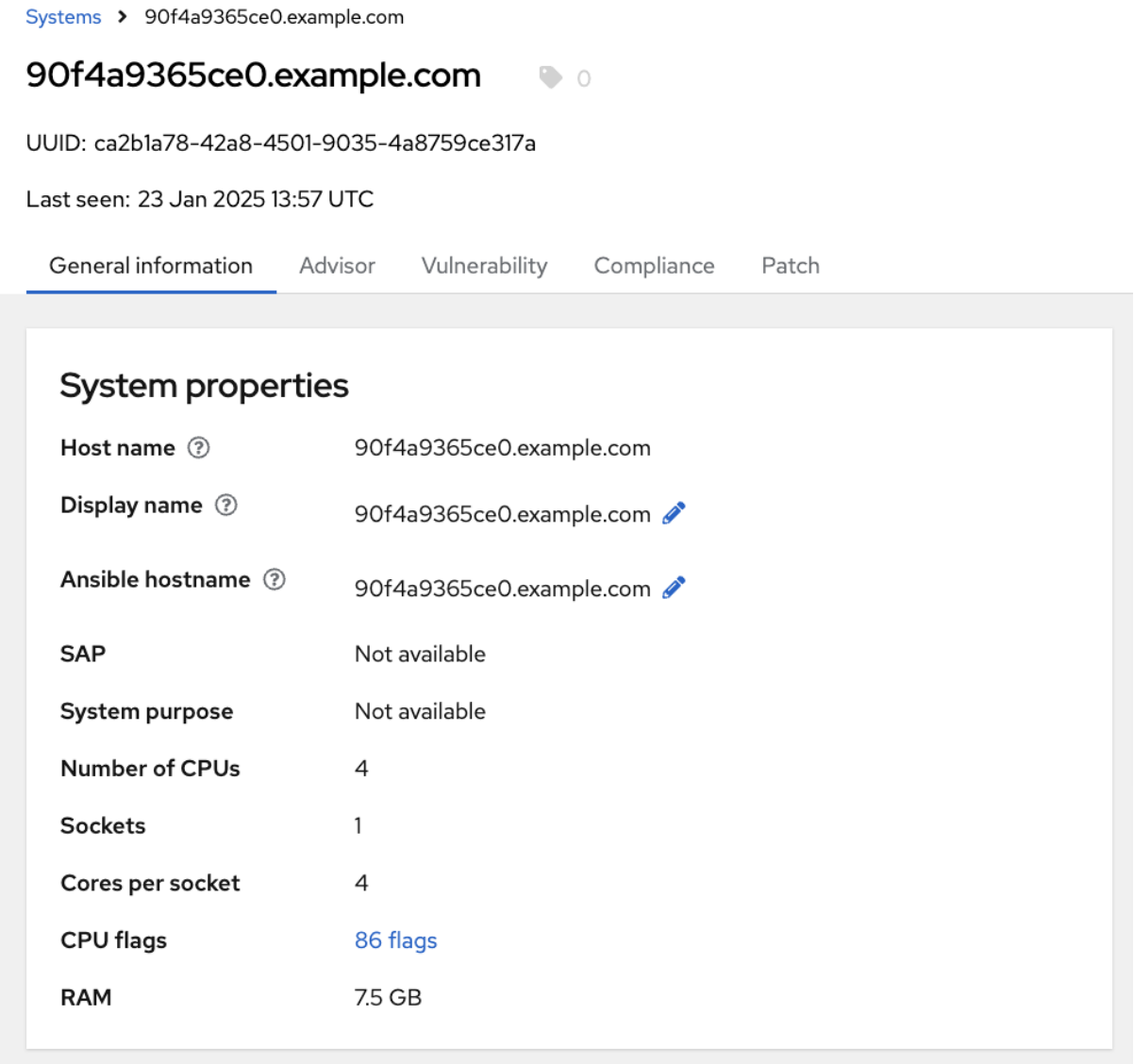
- When obfuscation is successfully enabled, the hostname gets masked in the Red Hat Lightspeed console UI, logs, and any archive data files that Red Hat collects.
- If you configure hostname obfuscation on more than one system, you might see multiple systems with the same hostname in the Red Hat Lightspeed GUI as a result of obfuscation.
- After you enable obfuscation, there might be instances where the original hostname displays in the command-line output of some **insights-client** commands.
- The original hostname of the system in **/etc/hostname**:

```
RTP.data.center.01
```

- Obfuscated **/etc/hostname** as displayed in Red Hat Lightspeed:

```
90f4a9365ce0.example.com
```

- Obfuscated hostname and IP address in the Red Hat Hybrid Cloud Console:



5.2.7. Obfuscation in Red Hat Satellite

For Red Hat Satellite-managed systems, obfuscation must be enabled in the Satellite web interface before configuring clients to prevent double-obfuscation.

Additional resources

- Red Hat Cloud settings

5.3. DATA REDACTION

You can configure data redaction to remove specific content such as files, command output, or keyword patterns from the collected archive. Data redaction can be used to exclude data containing personally identifiable information (PII).

5.3.1. Data redaction considerations and limitations

To prevent hostname, IP, and MAC addresses from being collected or sent to Red Hat, implement the obfuscation feature in combination with a Red Hat Satellite Server in your environment.



IMPORTANT

Directly connected systems. You cannot achieve true obfuscation or redaction of system information. The subscription-manager client must remain enabled and continues to report data from the system.

This applies regardless of how insights-client is configured.

You cannot prevent the collection of certain key system information such as the hostname, IP, and MAC addresses, and having that information sent to Red Hat. For directly connected systems, even configuring data obfuscation within the insights-client does not achieve the expected results.

Systems connected to a Red Hat Satellite or Capsule. You cannot fully prevent collection of key system information such as hostname, IP, and MAC addresses. However, you can adjust the feature level and configure Satellite so that data is never sent to Red Hat.

5.3.2. Data collection rules and redaction

Manage data collection and redaction by using YAML configuration files and the **insights-core** datasource catalog.

Additional resources

- Datasources
- Displaying the client version
- YAML-style denylist configuration for insights-client(Red Hat Knowledgebase)
- The **insights-core** datasource catalog
- System Information Collected by Red Hat Lightspeed (Red Hat Knowledgebase)
- Red Hat Insights Client Core Collection Description (Red Hat Knowledgebase)
- Red Hat Lightspeed data and application security

5.3.3. YAML files for redaction

To redact data in Red Hat Lightspeed, you need insights-client 3.0 and the following YAML configuration files to control the redaction actions:

- **file-redaction.yaml**
- **file-content-redaction.yaml**

You can use one or both files, depending on the content you want to redact.

To find the items to redact, insights-client uses the default configuration of the **insights-client.conf** configuration file to call the **file-redaction.yaml** and **file-content-redaction.yaml** files. The following example shows the default redaction configuration in the **insights-client.conf** file:

```
# Location of the redaction file for commands, files, and components
#redaction_file=/etc/insights-client/file-redaction.yaml

# Location of the redaction file for patterns and keywords
#content_redaction_file=/etc/insights-client/file-content-redaction.yaml
```

You do not need to change the configuration of the **insights-client.conf** file, but you do need to create the YAML files.

/etc/insights-client/file-redaction.yaml

Lists commands and files that you want to redact. A Python data cleaning process runs on the **file-redaction.yaml** file and redacts the listed commands and files. When the Python data cleaning process runs, it redacts the specified content before adding it to the archive file.

/etc/insights-client/file-content-redaction.yaml

Defines pattern redaction and keyword replacement. For pattern redaction, the process redacts patterns or regular expressions that match those specified in the YAML file. For keyword replacement, the process replaces the specified keywords with generic identifiers.

5.3.4. Configure file redaction

You can create the **/etc/insights-client/file-redaction.yaml** file and include a list of commands and system files that you want redacted. When the data redaction takes place, a Python data cleaning process runs, and analyzes the contents of the YAML file.



NOTE

The output of the listed commands or files does not get included in the uploaded archive file.

Prerequisites

- You must be familiar with the basics of YAML syntax. For more information about YAML, see yaml.org.
- You must have root-level access to the system.

Procedure

1. Use an editor to create the **/etc/insights-client/file-redaction.yaml** file.
2. Enter the strings, **files:** and **commands:**, on separate lines in the YAML file.

```
files:

commands:
```

3. Specify the files and commands you want to redact:

a. On the line following **files:**, enter the files that you want to redact. Use the information in the Datasources catalog to identify which files and commands to specify. For example, if you want to redact the **auditd.conf** file, this is how it would look:

```
files:
  - /etc/audit/auditd.conf
```

b. On the line following **commands:**, enter the commands that you want to redact. Use the information in the Datasources catalog to identify which commands to specify. For example, if you want to redact the **ethtool -i** command, this is how it would look:

```
commands:
  - ethtool_i
```

4. Save the YAML file in **/etc/insights-client/**.

5. On the command line, run **ll file-redaction.yaml** as root to verify that the **file-redaction.yaml** file permissions are **root** owner only.

```
[root@rhlightspeed]# ll file-redaction.yaml
-rw-------. 1 root root 145 Sep 25 17:39 file-redaction.yaml
```

The following example shows a sample **file-redaction.yaml** file that includes commands and files to redact. Comments, which are lines preceded by a hash symbol (#), also offer guidance to help you configure the YAML file.

```
# file-redaction.yaml
---
# Redact the entire output of commands
# Specify commands with the full command or with a "symbolic_name" such as “ethtool_i.”
# Refer to the “Datasource Catalog” and “General Datasources” at https://insights-core.readthedocs.io/en/latest/specs_catalog.html#general-datasource for a full list of available symbolic_names, and the commands and files they correspond to.

 commands:
- /bin/rpm -qa
- /bin/ls
- ethtool_i

# Redact the entire output of files
# Specify files either by full filename or
by the "symbolic_name" for example, “cluster_conf.”
# Refer to the “Datasource Catalog” and “General Datasources” at https://insights-core.readthedocs.io/en/latest/specs_catalog.html#general-datasource for a full list of available symbolic_names, and the commands and files they correspond to.

files:
- /etc/audit/auditd.conf
- cluster_conf
```

Verification

To verify that your redaction file is working, you can run the **insights-client** command with the **--no-upload** option, then review the output messages on your console or terminal.

1. On the command line, enter the **insights-client** command with the **--no-upload** option, and press **Return**.

```
[root@rhlightspeed]# insights-client --no-upload
```

2. The command runs and displays informational messages. The following example shows the redaction of the **dmesg** command and the **cluster.conf** file.

```
WARNING: Excluding data from files
Starting to collect Red Hat Lightspeed data for ITC-4
WARNING: Skipping command /bin/dmesg
WARNING: Skipping file /etc/cluster/cluster.conf
Archive saved at /var/tmp/qsINM9/rhlightspeed-ITC-4-20190925180232.tar.gz
```

The generated archive file gets saved to **/var/tmp** but the file is not uploaded to Red Hat.

5.3.5. Configure YAML pattern and keyword redaction

Configure the **file-content-redaction.yaml** file to redact data by using pattern redaction and keyword replacement methods.

Pattern redaction uses either a pattern match or regular expression match. In keyword replacement, a Python data cleaning process replaces the keyword with a generic identifier.

Prerequisites

- You must be familiar with the basics of YAML syntax. Explaining YAML is beyond the scope of this procedure.
- You have root-level access to the system.

Procedure

1. Use an editor to create the **/etc/insights-client/file-content-redaction.yaml** file.

```
# file-content-redaction.yaml
---
# Pattern redaction per matching line
#  Lines that match a pattern are excluded from files and command output.
#  Patterns are processed in the order that they are listed.
# Example

patterns:
 - "a_string_1"
 - "a_string_2"

# Regular expression pattern redaction per line
#  Use "regex:" to wrap patterns with regular expressions"
# Example

patterns:
 regex:
 - "abc.*def"
 - "localhost[[:digit:]]"
```

```
# Keyword replacement redaction
#  Replace keywords in files and command output with generic identifiers
#  Keyword does not support regex
# Example

keywords:
- "1.1.1.1"
- "My Name"
- "a_name"
```

2. Ensure the **file-content-redaction.yaml** file permissions are set for **root** owner only.

```
[root@rhlightspeed]# ll file-content-redaction.yaml
-rw-------. 1 root root 145 Sep 25 17:39 file-content-redaction.yaml
```

5.4. THE INSIGHTS-CLIENT ARCHIVE

To confirm what data the archive file sends to Red Hat, inspect the archive file. If you use obfuscation or redaction, you can inspect the archive before it uploads. If you want to preserve the archive file, you can keep it on your system.

5.4.1. Verify the archive before uploading

Use **insights-client --no-upload** to save the archive locally. You can view the information that the client sends to Red Hat Lightspeed, and verify your obfuscation or redaction settings.

Prerequisites

- You have a root user access to your system

Procedure

1. Enter the **insights-client** command with the **--no-upload** option.

```
[root@rhlightspeed]# insights-client --no-upload
```

The command displays informational messages when redaction or obfuscation is applied.

```
WARNING: Excluding data from files
Starting to collect Red Hat Lightspeed data for ITC-4
WARNING: Skipping patterns found in remove.conf
WARNING: Skipping command /bin/dmesg
WARNING: Skipping command /bin/hostname
WARNING: Skipping file /etc/cluster/cluster.conf
WARNING: Skipping file /etc/hosts
Archive saved at /var/tmp/qsINM9/rhlightspeed-ITC-4-20190925180232.tar.gz
```

2. Navigate to the temporary storage directory as shown in the *Archive saved at* message.

```
[root@rhlightspeed]# cd /var/tmp/qsINM9/
```

3. Unpack the compressed **tar.gz** file.

```
[root@rhlightspeed]# tar -xzf rhlightspeed-ITC-4-20190925180232.tar.gz
```

The **tar** command extracts the archive contents into a new directory.

5.4.2. Verify the archive after uploading

After the archive uploads to Red Hat Lightspeed, use the **--keep-archive** option to retain a local copy for inspection. You can then verify your obfuscation or redaction settings.

Prerequisites

- You have root user access to your system

Procedure

1. Enter the **insights-client** command with the **--keep-archive** option as a root user:

```
[root@rhlightspeed]# insights-client --keep-archive
```

The command displays informational messages.

```
Starting to collect Red Hat Lightspeed data for ITC-4
Uploading Red Hat Lightspeed data.
Successfully uploaded report from ITC-4 to account 6229994.
Red Hat Lightspeed archive retained in /var/tmp/ozM8bY/rhlightspeed-ITC-4-
20190925181622.tar.gz
```

2. Navigate to the temporary storage directory displayed in the **Red Hat Lightspeed archive retained in** message.

```
[root@rhlightspeed]# cd /var/tmp/ozM8bY/
```

3. Unpack the compressed **tar.gz** file.

```
[root@rhlightspeed]# tar -xzf rhlightspeed-ITC-4-20190925181622.tar.gz
```

The **tar** command extracts the archive contents into a new directory.

CHAPTER 6. ORGANIZE SYSTEMS IN RED HAT LIGHTSPEED BY USING FILTERING AND TAGS

Filter and organize your systems in Red Hat Lightspeed by using SAP workload tags, Satellite host group tags, and custom tags. Tagging enables you to focus on specific system groups across multiple services, which simplifies management for large-scale deployments.

6.1. FILTERS AND TAGS FOR SAP WORKLOADS

Red Hat Lightspeed automatically tags systems by SAP workload type and SAP ID (SID), which you can use to filter and organize SAP systems.

Use these automatic tags to filter and organize your SAP systems throughout the Red Hat Lightspeed application. No customization or manual tagging is required. To filter to view your SAP workloads, use **Workloads > SAP**.

6.2. FILTERS AND TAGS FOR SATELLITE HOST GROUPS

Satellite host groups are configured in Satellite and are automatically recognized by Red Hat Lightspeed.

6.3. CUSTOM SYSTEM TAGGING

Apply custom tags to your registered systems to add contextual markers and organize your Red Hat Lightspeed inventory. With custom tags, you can filter and focus on related groups of systems, which simplifies management for large-scale deployments.

You can also apply predefined tags to your systems. The advisor service uses predefined tags to generate targeted recommendations for systems that require specific attention, such as those with high security needs.

6.3.1. Filter structure

Filters use a **namespace=value** or **key=value** paired structure.

- **Namespace.** The namespace is the name of the ingestion point, *insights-client*. This value cannot be changed. The **tags.yaml** file is abstracted from the namespace, which is injected by the client before upload.
- **Key.** You can create the key or use a predefined key from the system. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.
- **Value.** You can define your own descriptive string value. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.

6.3.2. Create a custom group and the tags.yaml file

To create and add tags to **/etc/insights-client/tags.yaml**, use **insights-client** with the **--group=<name-you-choose>** option. Running this command on your systems that are registered to Red Hat Lightspeed completes the following actions:

- Creates the **etc/insights-client/tags.yaml** file

- Adds the **group=** key and **<name-you-choose>** value to **tags.yaml**
- Uploads a fresh archive from the system to the Red Hat Lightspeed service, making the new tag immediately visible along with your latest results

Prerequisites

- Root-level access to your system.

Procedure

1. Run the following command as root, adding your custom group name in place of **<name-you-choose>**:

```
[root@server ~]# insights-client --group=<name-you-choose>
```

2. Optional: To add additional tags, edit the **/etc/insights-client/tags.yaml** file.
3. Navigate to **Inventory → Systems** and log in if necessary.
4. Click the **Filter by tags** drop-down menu. You can also use the search box to enter all or part of the tag's name to automatically show systems with that text in the tags.
5. Scroll up or down the list to locate the tag.
6. Click the tag to filter by it.
7. Verify that your system is among the results on the advisor systems list.
   a. Navigate to **Inventory → Systems** and log in if necessary.
   b. Activate the **Name** filter and begin typing the system name until you see your system, then select it.
   c. The tag symbol is a darker color, and the number beside it shows the correct number of tags applied.

6.3.3. Add and modify custom system tags by editing tags.yaml

Edit the **tags.yaml** file to add or modify custom system tags after creating a **group** tag. Using custom system tags can help you streamline the organization and filtering of systems when you use Red Hat Lightspeed inventory and other services.

Prerequisites

- Root-level access to your system.

Procedure

1. Open the tag configuration file, **tags.yaml**, in an editor.

```
[root@server ~]# vim /etc/insights-client/tags.yaml
```

2. Edit the file contents or add **key=value** pairs as needed. Use a mix of capitalization, letters, numbers, symbols, and whitespace. Example **tags.yaml** with multiple tags:

```
group: _group-name-value_
location: _location-name-value_
description:
- RHEL8
- SAP
key 4: value
```

3. Save your changes and close the editor.

4. Initiate an upload to Red Hat Lightspeed.

```
[root@server ~]# insights-client
```

5. Navigate to **Inventory → Systems** and log in if necessary.

6. In the **Filter Results** box, click the down arrow and select one of the filters or enter the name of the filter and select it.



NOTE

You can search by the tag key or by its value.

7. Find your system among the results.

8. Verify that the filter icon is highlighted and shows a number representing the number of filters applied to the system.

6.4. PREDEFINED TAGS

In addition to custom tags, Red Hat Lightspeed provides predefined tags that the advisor service can use to create targeted recommendations. These predefined tags are useful for systems that require more security or need different networking performance levels.

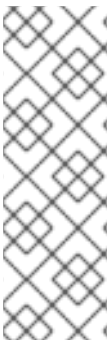
To get the extended security hardening and enhanced detection and remediation capabilities offered by predefined tags, opt in and configure the tags. After configuration, the advisor service provides recommendations based on tailored severity levels and the preferred network performance that applies to your systems.

Use the **/etc/insights-client/tags.yaml** file to tag systems with predefined tags, similar to how you tag systems in the inventory service. The predefined tags are configured using the same **key=value** structure used to create custom tags. Details about the Red Hat-predefined tags are in the following table:

Table 6.1. List of supported predefined tags

Key	Value	Note

Key	Value	Note
security	**normal** (default) / **strict**	With the **normal** (default) value, the advisor service compares the system's risk profile to a baseline that comes from the default configuration of the most recent version of RHEL and from often-used usage patterns. This option keeps recommendations focused, actionable, and low in numbers. The **strict** value is best for systems that are security-sensitive. This option uses a stricter baseline and can show recommendations even on the most up-to-date RHEL installations.
network_performance	**null** (default) / **latency** / **throughput**	The network performance values, either latency or throughput according to your business requirement, affect the severity of an advisor service recommendation to a system.



NOTE

The names of predefined tag keys are reserved for specific use. If you already use the key **security**, with a value that differs from one of the predefined values, you do not see a change in your recommendations. You only see a change in recommendations if your existing **key=value** is the same as one of the predefined keys. For example, if you have a **key=value** of **security: high**, your recommendations do not change because of the predefined tags that Red Hat has already defined. If you currently have a **key=value** pair of **security: strict**, you see a change in the recommendations for your systems.

Additional resources

- Using system tags to enable extended security hardening recommendations
- Leverage tags to make Red Hat Lightspeed Advisor recommendations understand your environment better
- Custom system tagging

6.4.1. Configure predefined tags

Configure predefined tags to adjust recommendations and gain extended security hardening, enhanced detection, and remediation capabilities within the Red Hat Lightspeed advisor service.

Prerequisites

- You have root-level access to your system.
- You have insights-client installed.
- You have systems registered within the insights-client.
- You have created the **tags.yaml** file. For information about creating the **tags.yaml** file, see Configure custom group tags.

Procedure

1. Use the command line and your preferred editor to open **/etc/insights-client/tags.yaml**. (The following example uses Vim.)

```
[root@server ~]# vi /etc/insights-client/tags.yaml
```

2. Edit the **/etc/insights-client/tags.yaml** file to add the predefined **key=value** pair for the tags. This example shows how to add **security: strict** and **network_performance: latency** tags.

```
# cat /etc/insights-client/tags.yaml
group: redhat
location: Brisbane/Australia
description:
- RHEL8
- SAP
security: strict
network_performance: latency
```

3. Save your changes.
4. Close the editor.
5. Optional: Run the **insights-client** command to generate an upload to Red Hat Lightspeed, or wait until the next scheduled Red Hat Lightspeed upload.

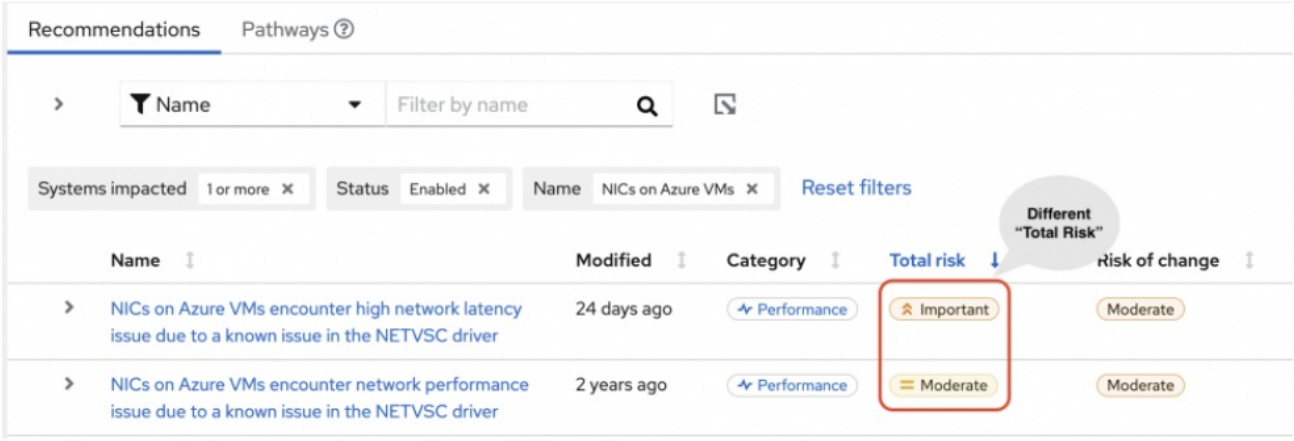
```
[root@server ~]# insights-client
```

Verification

After generating an upload to Red Hat Lightspeed (or waiting for the next scheduled Red Hat Lightspeed upload), you can discover whether the tags are in the production environment by accessing Red Hat Lightspeed > RHEL > Inventory > Systems . Find your system and look for the newly created tags. A table displays:

- Name
- Value
- Tag Source (for example, insights-client).

The following image of the advisor service shows a system with the **network_performance: latency** tag configured.



The system shows a recommendation with a higher Total Risk level of Important. The system without the **network_performance: latency** tag has a Total Risk of Moderate. You can make decisions about prioritizing the system with higher Total Risk.

6.5. ADDITIONAL RESOURCES

- Refining your view of systems in inventory
- Enable SQL Server recommendations in Advisor
- Group and tag filters in the compliance service

CHAPTER 7. TROUBLESHOOT ISSUES AND FIND REFERENCE INFORMATION

Find information to diagnose common connection failures and apply specific software fixes to maintain system health and connectivity.

7.1. REAL-TIME SCHEDULING CPU CONFLICT

The insights-client executes commands that collect data on your system. It has a configuration restriction that limits its CPU usage to no more than 30%.

You can define this restriction in the configuration file with the following option:

insights-client-boot.service: CPUQuota=30%

This configuration prevents the insights-client from creating a CPU spike that could interfere with other applications on your system, specifically applications that depend on real-time scheduling.

If you need to enable real-time scheduling, you can disable the CPU quota restriction. The risk of removing this configuration is minimal. However, CPU usage can become high when the insights-client runs. If high CPU usage impacts other services on your system, contact Red Hat support.

7.2. NETWORK CONNECTIVITY OF REQUIRED PORTS

If you experience communication failures, it might be because your system has a blocked port interfering with communication. To troubleshoot, verify that your system can reach all required Red Hat service destinations and TCP ports.

Check connectivity

Ensure that all required ports are open and destinations are accessible. The system needs access for the Subscription Manager and the insights-client to run.

Check the list ofTCP ports and destinations

Review the Red Hat Satellite documentation for networking requirements and port configuration.

Additional resources

- Red Hat Satellite documentation

7.3. CREATE A DIAGNOSTIC LOG FOR SUPPORT

You can create a diagnostic log to share with the support team.

Prerequisites

- You have root user access to your system.
- You have installed the insights-client on your system.

Procedure

1. Enter the **insights-client** command with the **--support** option.

```
[root@rhlightspeed]# insights-client --support
```

The command displays informational messages while creating the support file.

```
Collecting logs...
Red Hat Lightspeed version: insights-core-3.0.121-1
Registration check:
status: True
unreachable: False
. . . .
Copying Red Hat Lightspeed logs to archive...
Support information collected in /var/tmp/H_Y43a/insights-client-logs-20190927144011.tar.gz
```

2. Navigate to the collection directory as shown in the *Support information collected in* message.

```
[root@rhlightspeed]# cd /var/tmp/H_Y43a
```

3. Unpack the compressed **tar.gz** file.

```
[root@rhlightspeed]# tar -xzf insights-client-logs-20250912112824.tar.gz
```

Extracting the **tar.gz** copies the log files into your current directory. You can share the **tar.gz** file with the support team if requested.

CHAPTER 8. CLIENT COMMAND-LINE OPTIONS AND CONFIGURATION DETAILS

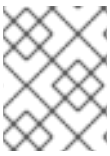
You can reference command-line options, configuration parameters, and debug options for insights-client to troubleshoot issues and customize client behavior. Use this reference to verify command syntax, configure insights-client settings, and enable debugging.

8.1. OPTIONS FOR INSIGHTS-CLIENT COMMAND

As a system administrator with root privileges, you can use the **insights-client** command and its options to control the insights-client operation on your system.

Updates to the **insights-client.rpm** occur less often than updates to individual components in Red Hat Lightspeed. The frequency of updates could cause the man page to not include the most recent information about **insights-client**.

Each time you enter the **insights-client** command, the client collects data and sends it to Red Hat Lightspeed.



NOTE

Using the **insights-client --display-name** command to set the display name takes effect immediately, but does not run the insights-client.

Table 8.1. insights-client user command options

Option	Description
--help **-h**	Display help information
--register	Register the host to Red Hat Lightspeed by using the information in **/etc/hostname**. Enables the nightly cron job unless **--disable-schedule** is set.
--unregister	Unregister the host from Red Hat Lightspeed.
--display-name=*DISPLAY_NAME*	Set or change the host display name in the GUI. Use it with the **--register** option to set the value of **display_name** during the host registration if you want to use a different name than the one specified in **/etc/hostname**. You can also use the **--display-name** option after registration to override the name shown for this host in Red Hat Lightspeed inventory.
--group=*GROUP*	Add a host to a *GROUP* during registration. Define Group names in **/etc/insights-client/tags.yaml**.

Option	Description
--retry=*RETRIES*	Set the number of times to retry an upload. The default is 1. The retry interval is 180 seconds, which is how long the insights-client waits until retrying the upload. NOTE: In the scheduler, the number of retries is 3.
--validate	Validate **/etc/insights-client/tags.yaml**, **/etc/insights-client/file-redaction.yaml**, and **/etc/insights-client/file-content-redaction.yaml** files.
--quiet	Log only error messages to the terminal output.
--silent	Disable message logging to the terminal.
--enable-schedule	Enable the job schedule. By default, the insights-client runs daily, at or near midnight.
--disable-schedule	Disable the nightly job schedule.
--conf=*CONF* **-c=*CONF***	Use a custom configuration file by specifying a file path for *CONF* instead of using the default **/etc/insights-client/insights-client.conf** file. IMPORTANT: This option is not available for RHEL 10 systems.
--no-upload	Runs the insights-client without uploading the archive to Red Hat Lightspeed. The archive is stored in the **/var/tmp/** directory. When the command completes running, the file name displays.
--offline	Run the client without using network functionality. Using this command does not upload data to Red Hat Lightspeed, similar to using the **--no-upload** option.
--logging-file=*LOGFILE*	Write the log data to the specified log file. The default log file is **/var/log/insights-client/insights-client.log**. IMPORTANT: This option is not available for RHEL 10 systems.

Option	Description
--diagnosis	Get diagnostic information from the API. The system must be registered and uploaded at least once to use the **--diagnosis** option.
--compliance	Scan the system with OpenSCAP and upload the report to the Red Hat Lightspeed compliance service in the Red Hat Hybrid Cloud Console.
--compliance-policies	List the policies available to assign to the system. The output also provides information for the following categories: **Assigned** Lists the status of the policy as **False** (system does not have the policy assigned) or **True** (system has the policy assigned). **ID** Lists the unique ID (UUID) for the policy. **Title** Lists the name of the Security Content Automation Protocol (SCAP) policy to which you can add your system.
--compliance-assign	Assign the system to the policy by using the ID from the **insights-client --compliance-policies** output.
--compliance-unassign	Remove the system from the policy by using the ID from the **insights-client --compliance-policies** output.
--payload=*PAYLOAD*	Upload a specific archive payload file to Red Hat Lightspeed. You must set **--content-type**.
--content-type=*TYPE*	Set the content type for the **PAYLOAD** option. The *TYPE* value can be **gz**, **bz2**, **xz**, or **none**. The value of *TYPE* must match the **--compressor** used with the payload file.
--show-results	Display analysis results by using the **--check-results** option.
--output-dir=*DIR*	Write the collection to a specified directory instead of uploading to Red Hat Lightspeed.

Option	Description
--output-file=*FILE*	Write the collection to a specified archive instead of uploading to Red Hat Lightspeed.
--ansible-host=*ANSIBLE-HOST*	Set or change the Ansible hostname in Red Hat Lightspeed inventory. Run this during registration to specify a different hostname from the one listed in **/etc/hostname** or the configuration file. This also overrides the hostname that Red Hat Lightspeed remediations injects into generated playbooks targeting the Ansible host.
--list-specs	Show **insights-client** collection specs.
--build-packagecache	Refresh the system package manager cache.
--manifest=*MANIFEST*	The specified manifest YAML file that defines what Red Hat Lightspeed Core should collect.
--checkin	Do a lightweight check-in instead of a full upload.
--collector=*APPLICATION*	Run the specified application and upload its generated archive. For example, specify: ● **--collector malware-detection** to use Red Hat Lightspeed to help detect the presence of malware. For more information, see the *Additional resources* section. ● **--collector compliance** to instruct Red Hat Lightspeed to scan the system with OpenSCAP and upload the report. You can get the same result by using the`--compliance` option.

Additionally, use the following **insights-client** command options when you need to debug insights-client operations:

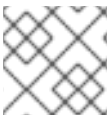
Table 8.2. **insights-client** debug options

Option	Description
--version	Print the versions of **insights-client** Client and Core.
--test-connection	Test connectivity to the Red Hat Lightspeed services.

Option	Description
--verbose	Log all debug output to the console.
--no-upload	Run the client, but do not upload the archive. The client stores the archive in the **/var/tmp/** directory. The file name displays when **insights-client** completes.
--keep-archive	Store the archive in the **/var/cache/insights-client/** directory after upload.
--support	Generate a diagnostic log for support.
--status	Display host registration status.
--net-debug	Log network calls to the console.

8.2. OPTIONS FOR THE INSIGHTS-CLIENT CONFIGURATION FILE

When the configuration file and the command line have similar options, the command-line option is executed when you enter the **insights-client** command. When the scheduler runs the client, the configuration file options are executed.



NOTE

You must enter the choices exactly as shown. **True** and **False** use initial capital letters.

The changes initiated by the options take effect either at the next scheduled run, or when you enter the **insights-client** command. Format the options as key=value pairs.

Table 8.3. insights-client.conf configuration options

Option	Description
ansible_host	Use this option if you want a different hostname when running Ansible playbooks.
authmethod=CERT	Set the authentication method. The default value is CERT.
auto_config=True	Use this to auto configure with Satellite Server. Values can be **True** (default) or **False.** NOTE: When **auto_config=True** (default), the authentication method used is **CERT**.

Option	Description
auto_update=True	Use this option to automatically update the dynamic configuration. The default value is **True.** Change the value to **False** if you do not want automatic updates.
base_url=cert-api.access.redhat.com:443/r/insights	Specify the base URL for the API.
cmd_timeout=120	Use this option to set the timeout limit for commands that run during collection. The value is measured in seconds. The command processes end when the timeout limit reaches the value of this option.
content_redaction_file	Omit lines or keywords from files and commands in the core collection. The core collection consists of a comprehensive set of data such as system logs, configuration files, and command outputs collected by the **insights-client**. You do not need to change the default configuration. The **content_redaction_file** option uses the **/etc/insights-client/file-content-redaction.yaml** file by default.
display_name	Set the display name for the **--display-name** option during registration. The default is to use **/etc/hostname**. NOTE: This value interacts with the **insights-client --display-name** command. If you use the command line to change the display name but the configuration file contains a different display name, the display name reverts to the configuration file value when the scheduler runs the insights-client.
http_timeout=120	Use this option to set the timeout limit for HTTP calls. The value is measured in seconds. The command stops the HTTP call when the timeout limit reaches the value.
[insights-client]	The configuration file must begin with this entry, even if you specify a custom path or name for the client configuration file.
loglevel=DEBUG	Use this option to change the log level. Values are: DEBUG, INFO, WARNING, ERROR, and CRITICAL. The default is DEBUG. The default log file location is **/var/log/insights-client/insights-client.log**.

Option	Description
obfuscation_list=<ipaddress_type>	Use this option to specify which IP connectivity protocol addresses to obfuscate in the data collection. By default, this value is not set and no data obfuscation occurs. Supported values are: **IPv4**, **IPv6**, **hostname**, and **MAC**. For example, to obfuscate the IP address (IPv6) and the hostname, set **obfuscation_list=ipv6,hostname**. IMPORTANT: Remove deprecated **obfuscate** and **obfuscate_hostname** options from the configuration file. If your configuration file contains both **obfuscation_list** and deprecated options, the **obfuscation_list** takes precedence and insights-client displays a warning message in the output.
proxy	Use this option to set the URL for your proxy. Example: http://user:pass@192.168.100.50:8080
redaction_file	Use this option to omit files or commands from the core collection. The core collection consists of a comprehensive result set. You do not need to change the default configuration. The **redaction_file** option uses **/etc/insights-client/file-redaction.yaml** by default.